DSA06 - Data Handling and Visualization | Activity 5: Normality Detection using Q-Q Plots | Slot D

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Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Aim

To identify whether the given datasets follow a normal distribution by observing Q-Q plots and comparing the sample quantiles with the theoretical quantiles.

Part A: Analysis of Q-Q Plots

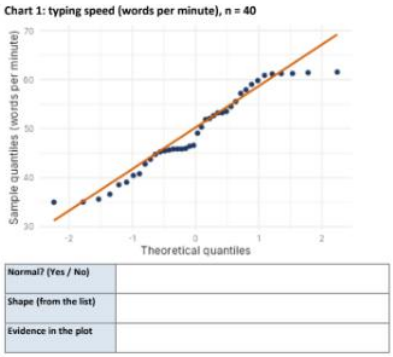


Chart 1: Typing Speed (n = 40)

- **Normality:** Yes, approximately normal.
- **Shape:** Normal.
- **Evidence:** Most points lie close to the reference line. Only small deviations are visible at the ends, indicating that the data approximately follows a normal distribution.

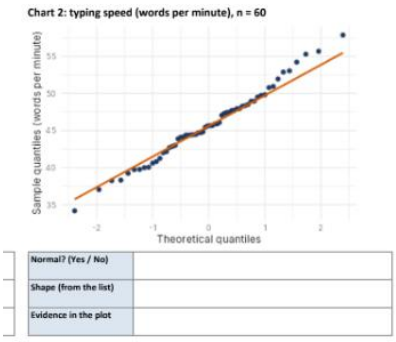
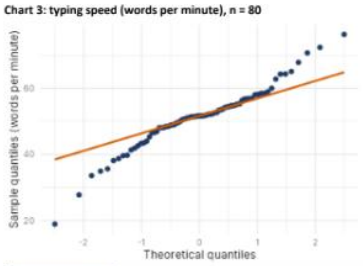


Chart 2: Typing Speed (n = 60)

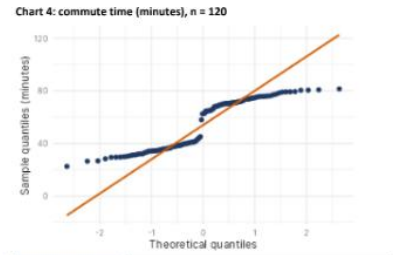
- **Normality:** No.
- **Shape:** Right-skewed.
- **Evidence:** The points deviate above the reference line in the upper tail, indicating positive skewness in the data.



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: Typing Speed (n = 80)

- **Normality:** No.
- **Shape:** Bimodal.
- **Evidence:** The points show a curved pattern and deviate from the reference line at both ends. This suggests that the data does not follow a normal distribution.



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 4: Commute Time (n = 120)

- **Normality:** No.
- **Shape:** Bimodal.
- **Evidence:** The points show a strong curved pattern with a sudden jump in the middle and flattening at the upper end. Therefore, the data is not normally distributed.

Part B (4 marks)

B1. Chart 3 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

Part B: Short Answers

B1. Chart 3 departs from normality. Name the departure and state the remedy you would try before using mean-and-standard-deviation methods.

Answer:

Chart 3 shows a departure from normality, with a pattern suggesting a bimodal distribution. If the data contains two different groups, I would separate the groups and analyze them individually. I would also consider robust statistical methods if the data still does not follow a normal distribution.

B2. Fill in the blanks to draw a Q-Q plot with its reference line in R using ggplot2.

Answer:

```
ggplot(df, aes(sample = value)) +
  stat_qq() +
  stat_qq_line()
```

Here, stat_qq() plots the sample quantiles against theoretical quantiles, and stat_qq_line() adds the reference line for checking normality.

Result

The four Q-Q plots were analyzed to identify normality and departures from a normal distribution. Chart 1 approximately follows a normal distribution, while Charts 2, 3, and 4 show deviations from normality. The analysis helps determine whether normality-based statistical methods are suitable for the given data.